

S4 · 15:45–16:00

Complex Genotype–Phenotype Correlations in Autism Spectrum Disorder: A Systems Biology Approach

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OVERVIEW – AUTISM SPECTRUM DISORDER (ASD)

ASD is a heterogeneous group of **neurodevelopmental disorders** (NDD) affecting **over 1%** of children worldwide.

It is defined by persistent challenges in social communication, sensory processing, as well as restricted and stereotyped patterns of behavior, interests and activities.



GENETIC MODEL ADDITIVE EFFECT

+ Rare variant

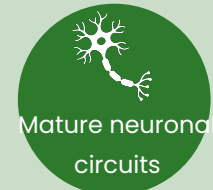
Single de novo or ultra-rare variants
→ chromatin remodeling complexes

Other genetic variants
→ synaptic strength & activity-dependent signaling

+ Common inherited background

→ Polygenic risk architecture

AFFECTED PROCESSES



Proteins involved converge on common pathways: synaptic plasticity, chromatin remodelers, transcription factors and the mTOR pathway.

How can we study the complex relationship between genotype and phenotype in ASD, using multiple biological data sources?



A Network Biology Framework for Patient Stratification through Mutational Load Assessment in Autism Spectrum Disorder

Develop a network biology framework integrating:

- genomic
- transcriptomic
- pathway data

to quantify **mutational load** and correlate with **phenotypic heterogeneity** in ASD.

PROJECT WORKFLOW

PATIENT-LEVEL AND PUBLICLY AVAILABLE DATA



COHORT OF 2,600 SIMPLEX FAMILIES



COHORT OF 415,000 PARTICIPANTS
INCLUDING OVER 175,000 PEOPLE WITH
ASD



SFARI

SIMONS FOUNDATION
AUTISM RESEARCH INITIATIVE

ASD-RELATED GENES



PPI INTERACTIONS



TISSUE-SPECIFICITY



PATHWAYS ANNOTATIONS

DATA COLLECTION

DATA ANALYSIS AND SINGLE-LAYER NETWORK CONSTRUCTION



DATA-TYPE DRIVEN ANALYSIS



MULTIPLEX IMPLEMENTATION



PROTEIN-PROTEIN INTERACTIONS
NETWORK

PATHWAY SIMILARITY NETWORK

GENE CO-EXPRESSION NETWORK



DATA INTEGRATION AND NETWORK

NETWORK PROPAGATION SIMULATIONS



VCF

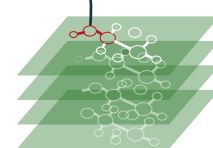
Vep

AlphaMissense

SpliceAI

Evo2

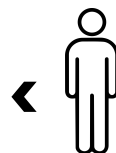
Multiplex Label
propagation
algorithms



Computing
PGS score



MUTATION
PERTURBED
MODULES



PHENOTYPIC DATA
INTEGRATION

MUTATIONAL LOAD ASSESSMENT

DATA COLLECTION

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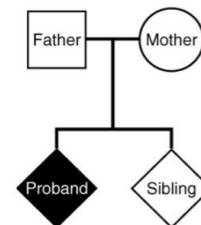
TISSUE- SPECIFICITY



PATHWAYS ANNOTATIONS

The Simons Simplex Collection (SSC) is a core project and resource of the Simons Foundation Autism Research Initiative (SFARI).

The SSC achieved its primary goal: establishing a permanent repository of genetic samples from simplex families and study the CNVs in these simplex families.



NEUROVIEW Volume 68, Issue 2, P182-195, October 21, 2010 · Open Archive · Download Full Issue

The Simons Simplex Collection: A Resource for Identification of Autism Genetic Risk Factors

Gerald D. Fischbach, A. J. B. · Catherine Lord

Article | Published: 29 October 2014

The contribution of *de novo* coding mutations to autism spectrum disorder

Ivan Iossifov, Brian J. O'Roak, Stephen J. Sanders, Michael Ronemus, Niklas Krumm, Dan Levy, Holly A. Stessman, Kali T. Witherspoon, Laura Vives, Karynne E. Patterson, Joshua D. Smith, Bryan Pasper, Deborah A. Nickerson, Jeanelle Dea, Shan Dong, Luis E. Gonzalez, Jeffrey D. Mandell, Shrikant M. Mane, Michael T. Murtha, Catherine A. Sullivan, Michael F. Walker, Zainulabedin Wapar, Liping Wei, A. Jeremy Wilsey, ... Michael Wigler

Nature 515, 216-221 (2014) | Cite this article

→ After approval, we obtained genomic data for 4,073 individuals;

→ Transcriptomic data for 3,979 autistic and nonautistic siblings;

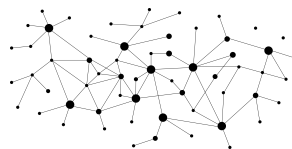
→ Phenotypic data for 11,901 individuals, including ADI-R, ADOS and other diagnostic questionnaires.

DATA INTEGRATION AND NETWORK MODELLING

DATA ANALYSIS AND SINGLE-LAYER NETWORK CONSTRUCTION



MULTIPLEX IMPLEMENTATION



NODES → proteins – genes – RNA

→ ID (ex. UNIPROT)

→ GENE NAME

→ NODE TYPE (SFARI OR NEIGHBOURS)

→ SFARI SCORE

→ MUTATIONS → IMEX ANNOTATIONS

ATTRIBUTES



EDGES

→ Weighted per layer

→ **PPI** – PHYSICAL INTERACTION · IMEx MI-score ≥ 0.4 (threshold applied to seed–neighbour edges)

→ **CO-EXPRESSION** – GTEx Brain cortex dataset · WGCNA biweight midcorrelation

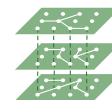
→ **PATHWAY SIMILARITY** – JACCARD / GENE-SET OVERLAP

→ **ATTRIBUTE**: EFFECT OF A MUTATION

MULTIPLEX INTEGRATION

Every layer encodes a different biological relationship.

→ Future: Multilayer integration



STATISTICAL VALIDATION: each network tested against Maslov–Sneppen degree-preserving null models.

PPI NETWORK

Workflow

Gene set from
SFARI (1,2,3 & 1S)



PPI interactions
extracted from IMEx



Filtering



Topological
Analysis



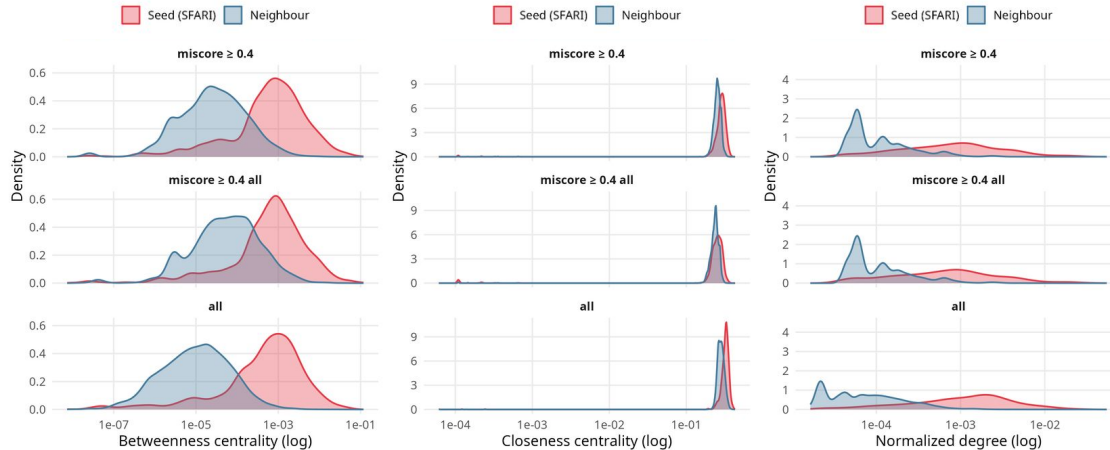
Statistical Validation



Random walk simulations

Centrality distributions — PPI network

Seed (SFARI ASD genes) vs first neighbours · nodes with value = 0 excluded (log scale)

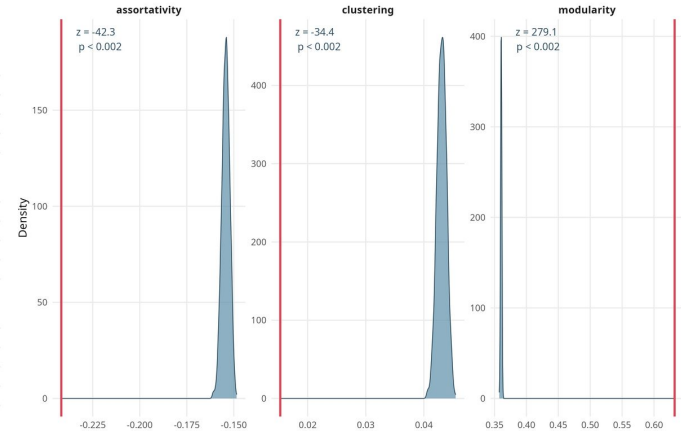


- $\text{mismatch} \geq 0.4$ threshold applied to seed-neighbour edges only · 9,228 nodes · 31,015 edges
- $\text{mismatch} \geq 0.4$ all threshold applied to all edges · 9,131 nodes · 27,863 edges
- all : no filtering · 16,423 nodes · 90,055 edges

← **Selected network**

Global topology metrics: observed vs null

1000 degree-preserving rewirings



PPI LABEL PROPAGATION

Label propagation were computed using two different algorithms implemented in DiffuStats.

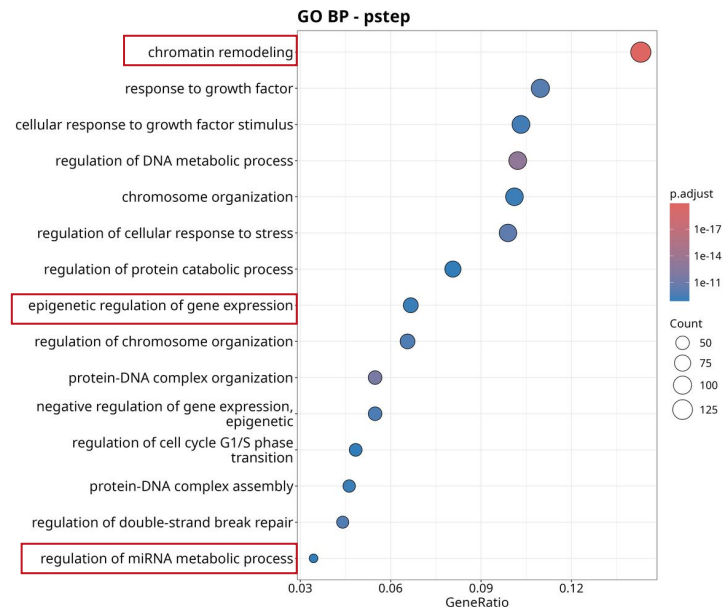
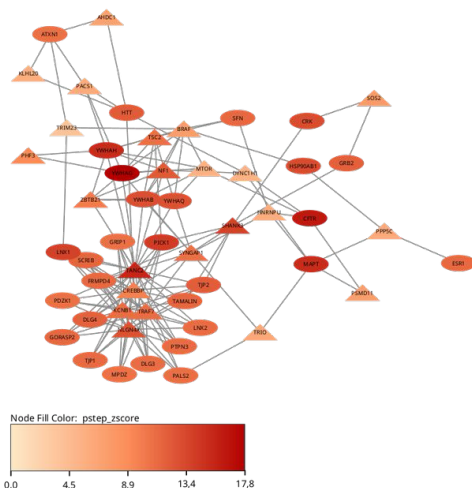
P-STEP RANDOM WALK: This kernel is more focused on local properties of the nodes. The kernel captures the local signal around SFARI seed genes, tight modules and direct partners.

Formula: $r(\lambda) = (\alpha - \lambda)^{-p}$, with $\alpha \geq 2$, $p \geq 1$

Initial signal: SFARI seed genes were initialized with a starting score proportional to their SFARI confidence category.

Score normalisation (mc). Diffusion scores were statistically normalised with the Monte Carlo method of diffuStats: the raw score is compared to a null distribution from input permutations on the labelled nodes, correcting for network structure.

Enrichment: ORA performed on the diffusion-reached neighbours with $FDR < 0.05$ (BH), ranked by z-score. Seeds excluded.



PPI LABEL PROPAGATION

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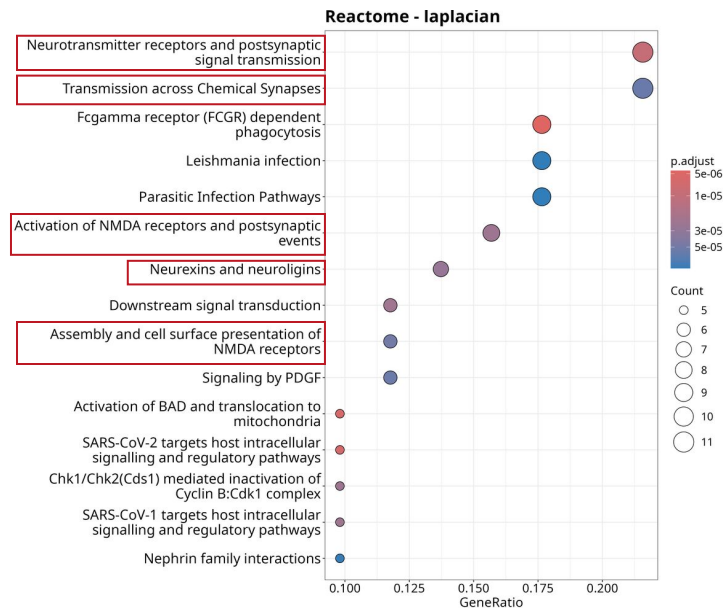
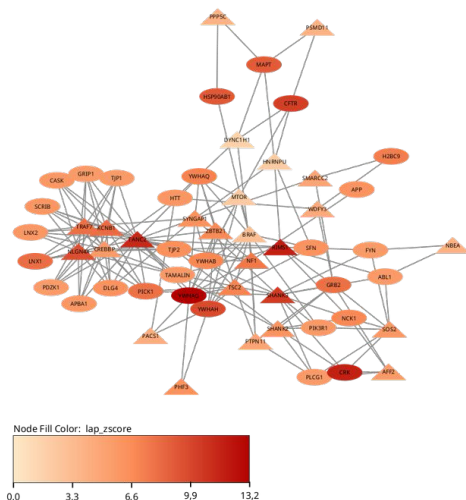
LAPLACIAN KERNEL: The regularised Laplacian kernel similar to the heat diffusion model, diffuses the signal more globally across the network, to capture functionally related genes beyond the immediate neighbourhood.

$$\text{Formula: } r(\lambda) = 1 + \sigma^2 \lambda$$

Initial signal: SFARI seed genes were initialized with a starting score proportional to their SFARI confidence category.

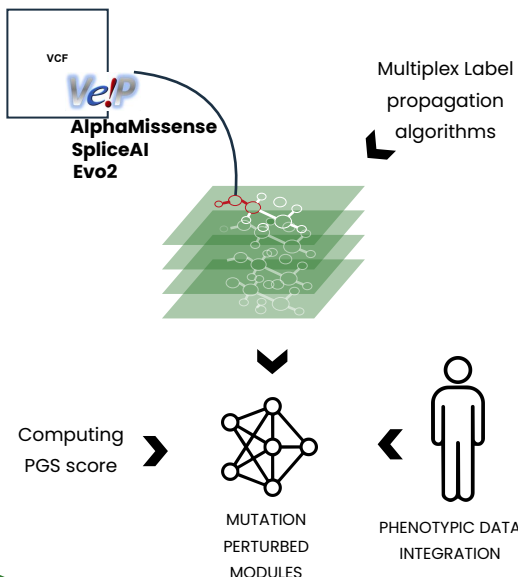
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MUTATIONAL LOAD ASSESSMENT

NETWORK PROPAGATION SIMULATIONS



1 VARIANT IMPACT ANNOTATION

Individual VCF → per-variant functional impact, mapped onto network nodes as individual mutational burden → one seed set per individual

ENSEMBL VEP**AlphaMissense****SpliceAI****Evo2****New!**

2 MULTIPLEX PROPAGATION

Random Walk with Restart on the multiplex-heterogeneous network, run independently for each individual → mutation-perturbed modules per patient.

3 INTEGRATED MUTATIONAL LOAD



Perturbed-module signal combined with per-individual **PGS** (SSC cohort)

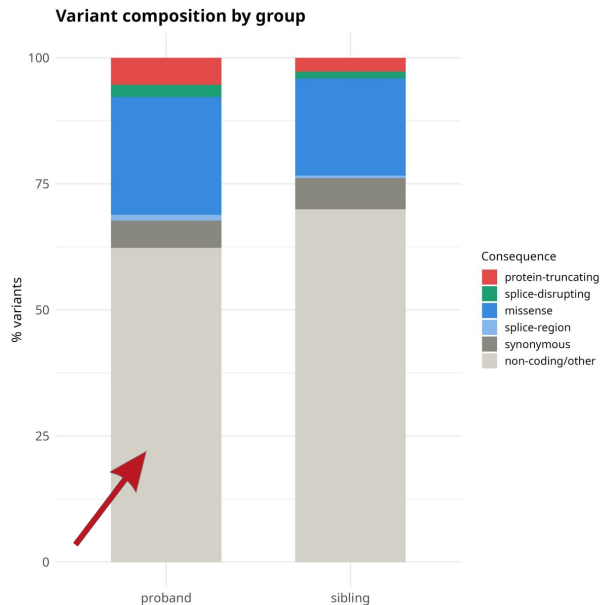
4 PHENOTYPIC CORRELATION & PATIENT STRATIFICATION

Proband-sibling design → correlate load + PGS with phenotypic features.
Stratify patients by phenotype.

VARIANT IMPACT ANNOTATION



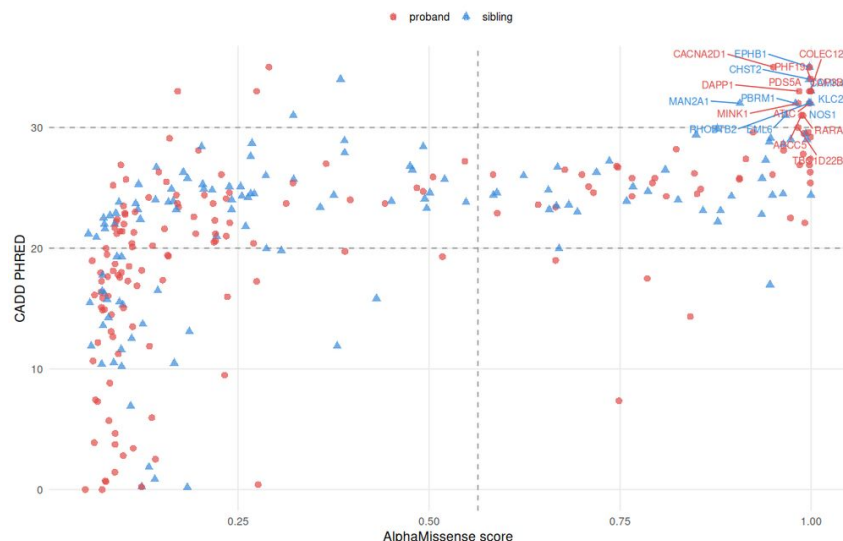
WORK IN PROGRESS



A substantial fraction of variants is **intronic**, beyond the reach of missense-focused tools. **Evo2** is added for non-coding effect prediction.

In collaboration with Prof. Raimondi and Dr. Natalia de Oliveira Rosa

CADD vs AlphaMissense



NEXT..

Each gene's variants are collapsed into a **single per-gene impact score**, e.g. the maximum predicted effect across its variants, weighted by zygosity. This score seeds the **patient-specific propagation**.

In collaboration with Prof. Raimondi and Dr. Natalia de Oliveira Rosa

CONCLUSIONS

TO WRAP UP...

- PPI layer → topological and enrichment analyses confirm convergence on synaptic, chromatin and signalling pathways. Label propagation with two kernels reveals complementary axes: chromatin and synaptic.
- Variant annotation completed with VEP, AlphaMissense, SpliceAI and Evo2(ongoing) → ready as input for the patient-specific propagation.

FUTURE PERSPECTIVES

- Finalise the multiplex framework (GTEx Brain Cortex co-expression network and pathway similarity)
- Patient-specific propagation weighted by variant impact
- Computing PGS for the individual in SSC cohort
- Correlation of perturbed modules with phenotypic features

THANK YOU FOR YOUR ATTENTION

Licata Group

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Marta Iannuccelli, MINT Senior
biocurator



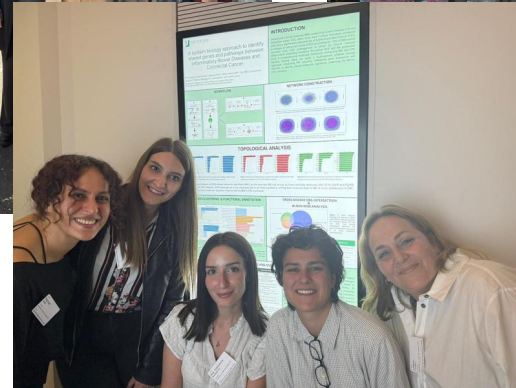
Bio@SNS

Prof. Francesco Raimondi, PhD

Dr. Natalia de Oliveira Rosa

All the families in the Simons

Simplex Collection and to SFARI.



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